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A Software Approach to the PPT2 Conjecture

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Intellectual Property Statement

Statement text regarding intellectual property goes here...

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Abstract

This thesis explores...

Povzetek

V tem delu raziskujemo...

Razširjeni povzetek

Daljši slovenski povzetek vsebine...

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Introduction

Quantum entanglement is one of the main nonclassical resources in quantum information theory. It underlies protocols for cryptography, communication, and distributed computation, and it also serves as a structural tool for understanding quantum channels [1]. A recurring practical question is how fast entanglement is degraded when channels are composed. This motivates studying classes of channels that preserve limited positivity properties but are not fully entanglement preserving.

Among these classes, positive-partial-transpose (PPT) channels play a central role. The PPT² conjecture asks whether the composition of two PPT channels is always entanglement breaking [2]. If true, this would identify a strong structural mechanism by which repeated noise destroys entanglement. If false, an explicit counterexample would reveal new high-dimensional behavior in the geometry of positive maps.

The conjecture is deeply connected with operational questions in quantum communication. Entanglement-breaking channels are the strongest form of noise for entanglement transmission: once such a channel is applied, no entanglement can survive between the channel output and any reference system. This directly constrains repeater-like settings and motivates the original interest in PPT² [3], [4].

Several important partial results are known. The conjecture is proven in low dimensions and for specific map families, including Choi-type maps [5], [6]. At the same time, it is widely expected that high-dimensional regimes exhibit more complicated behavior, where generic arguments from low dimensions no longer apply. This makes large-scale and carefully validated computation a relevant tool for progress.

This thesis adopts a software-first methodology. Instead of targeting a purely analytic proof, we build a reproducible computational framework that searches for high-dimensional counterexample candidates and evaluates them with two complementary mechanisms. The first mechanism is the standard DPS-based route for entanglement detection. The second mechanism is a polynomial/SOS route used to construct witness candidates with additional post-verification structure.

The methodological challenge is numerical reliability. In borderline cases, floating-point optimization can produce certificates that are suggestive but not mathematically robust. Consequently, a major part of this thesis is not only searching for candidates, but designing a protocol that separates robust evidence from numerical artifacts.

The main objectives are:

1. Design and implement a scalable Julia framework for generating PPT candidates, composing channels, and evaluating outputs.

2. Integrate DPS-style semidefinite tests with explicit logging and reproducibility support.
3. Add an SOS-based witness-generation route for cross-checking difficult instances.
4. Develop a reliability protocol based on tolerance sweeps, perturbation checks, and low-dimensional sanity controls.

The thesis contributions are primarily computational and empirical:

- an extensible workflow for PPT² experiments in Julia,
- a reproducible benchmark format for candidate states, witnesses, and solver settings,
- an implementation-level validation protocol that labels outcomes as validated, rejected, or inconclusive,
- a library of generated witness candidates and tested compositions, useful for later studies even if no final counterexample is found.

Even in the absence of a final counterexample, these contributions are valuable: they turn an informal numerical search into a repeatable scientific process and provide infrastructure for future analytical and computational work.

The rest of the thesis is organized as follows. Chapter 2 introduces the mathematical framework and notation. Chapter 3 presents the computational methods and software architecture. Chapter 4 reports experiments and diagnostics. Chapter 5 discusses limitations and reliability. Chapter 6 concludes with open problems and future directions.

Theoretical Background

This chapter formalizes the notation used throughout the thesis and introduces the core mathematical objects behind the computational pipeline.

PPT2 Conjecture

Let M_n denote complex $n \times n$ matrices, and let M_n^+ denote the cone of positive semidefinite matrices in M_n .

Definition 2.1 (Positive map). A linear map $\Phi : M_n \rightarrow M_m$ is positive if

$$X \in M_n^+ \implies \Phi(X) \in M_m^+.$$

following the standard map-theoretic setup used in [5], [7].

Definition 2.2 (k -positive and completely positive map). For $k \geq 1$, map Φ is k -positive if $\text{id}_k \circ \Phi$ is positive. Map Φ is completely positive (CP) if it is k -positive for all $k \geq 1$ [5], [7].

Definition 2.3 (Completely copositive map). Map Φ is completely copositive if $T \circ \Phi$ is completely positive, where T denotes matrix transposition [5], [7].

Definition 2.4 (Quantum channel). A linear map $\Phi : M_n \rightarrow M_m$ is a quantum channel if it is completely positive and trace preserving.

Scope note. Although channels are trace preserving by definition, the computational tests in this thesis are primarily positivity/sign based. Therefore, intermediate computations may use non-normalized maps, then normalize where needed for interpretation.

Definition 2.5 (PPT map). A map is PPT if it is both completely positive and completely copositive [5], [7].

Definition 2.6 (Entanglement-breaking map). Map $\Lambda : M_n \rightarrow M_m$ is entanglement breaking if for every bipartite state $\rho \in M_n \otimes M_m$, state $(\text{id}_n \otimes \Lambda)(\rho)$ is separable [5], [7].

Conjecture 2.7 (PPT² conjecture). If maps Φ and Ψ are PPT, then map $\Phi \circ \Psi$ is entanglement breaking [2].

Known results provide anchor points for this thesis. The conjecture holds in low-dimensional settings and for important structural classes such as Choi-type maps [5], [6]. These facts are used as calibration tests for our computational pipeline.

Concepts

Entanglement and Separability

Definition 2.8 (Bipartite quantum state). A bipartite state is matrix $\rho \in M_n \otimes M_m$ such that $\rho \geq 0$ and $\text{tr}(\rho) = 1$ [7].

Definition 2.9 (Separable state). State $\rho \in M_n \otimes M_m$ is separable if it can be written as

$$\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B,$$

where $p_i \geq 0$, $\sum_i p_i = 1$, and each ρ_i^A, ρ_i^B is a local density matrix. If state ρ is not separable, it is entangled [1], [7].

Definition 2.10 (Partial transpose and PPT state). Let $\Gamma_B := \text{id}_n \otimes T_m$, where T_m is transposition on subsystem M_m . State ρ is PPT if $\Gamma_B(\rho) \geq 0$ [7], [8], [9].

Proposition 2.11 (PPT criterion, necessity). If ρ is separable, then ρ is PPT [8], [9].

Remark 2.12. In low dimensions, PPT is also sufficient for separability; in higher dimensions it is not. Hence PPT entangled states exist [1].

Detecting Entanglement

Exact separability testing is computationally hard in general. Practical approaches therefore rely on convex relaxations and hierarchies.

Entanglement Witnesses

Definition 2.13 (Entanglement witness). Hermitian matrix W is an entanglement witness if:

1. for every separable state σ , $\text{tr}(W\sigma) \geq 0$,
2. there exists state ρ such that $\text{tr}(W\rho) < 0$.

This definition is consistent with the witness formulation used in [7].

Witnesses define one-sided certificates for entanglement and arise naturally as dual objects in convex optimization formulations of separability [1].

Interpretation used in this thesis. A valid negative witness value certifies entanglement of the tested state. The converse does not hold: failure to find a witness is not a certificate of separability.

DPS Hierarchy

Definition 2.14 (DPS hierarchy, informal). For each level $\ell \geq 1$, the DPS test defines an SDP-feasible set S_ℓ based on symmetric extensions and positivity constraints, with:

the feasible sets becoming tighter as level increases. In particular, if a state is outside set S_ℓ , it is certified entangled at level ℓ [7], [10].

The DPS framework yields a complete family of separability criteria in the limit and is a standard practical baseline for entanglement detection [10]. Refined hierarchy variants improve quantitative behavior in several regimes [11].

In this thesis, DPS serves as Pipeline A. Because solver tolerance can blur near-boundary instances, we never treat a single borderline SDP output as conclusive evidence.

Certification caveat. In exact arithmetic, detected witness negativity is one-sided evidence for entanglement. In floating-point arithmetic, tiny negative values can be numerical artifacts. This is why all near-boundary detections are passed to the reliability protocol before any claim is made.

Positive Maps and Choi Representation

Definition 2.15 (Choi matrix). For map $\Phi : M_n \rightarrow M_n$, its Choi matrix is

$$J(\Phi) = (\text{id}_n \circ \Phi)(\Omega\Omega^*),$$

where $\Omega = \sum_{i=1}^n e_i x e_i$.

Theorem 2.16 (Choi criterion). Map Φ is completely positive if and only if $J(\Phi) \geq 0$ [12].

Definition 2.17 (PPT condition in Choi form). Map Φ is PPT if both $J(\Phi) \geq 0$ and $\Gamma_{B(J(\Phi))} \geq 0$ hold.

The Choi correspondence allows direct conversion between map composition tests and matrix-level PSD/PPT checks.

SOS Viewpoint for Witness Construction

The second computational route in this thesis relies on polynomial positivity and SOS relaxations. The key idea is to build candidate objects that are positive on relevant domains but not representable in a way corresponding to complete positivity. This creates witness candidates that can complement DPS outputs in ambiguous cases [13], [14], [15].

From an optimization perspective, this turns witness construction into a sequence of SDPs involving Gram-matrix certificates. Although still numerical, this route provides additional structure for post-processing and validation.

Working Assumptions for Computation

To keep claims rigorous despite numerical optimization, we adopt the following policy throughout the thesis:

1. A candidate is never accepted as a counterexample from one solver run alone.
2. Results in dimensions where PPT² is proven are used as mandatory control tests.
3. Near-zero certificates are labeled inconclusive unless stable under tolerance and perturbation sweeps.
4. Agreement between independent pipelines is treated as stronger evidence than agreement within one pipeline.

Methods

This chapter defines the computational pipeline used to search for counterexamples to PPT² and to evaluate their reliability.

Chapter Goals and Design Constraints

The methodology is designed around three constraints:

1. Scalability: candidate generation and testing must run in high dimensions with large sample counts.
2. Redundancy: at least two independent detection routes are used on hard instances.
3. Auditability: every run is logged so results can be reconstructed from seed and solver configuration.

The final system combines two pipelines:

- Pipeline A: randomized PPT candidate generation followed by DPS-based detection,
- Pipeline B: polynomial/SOS witness construction and independent witness testing.

Computational Inputs and Outputs

Input parameters of an experiment batch are:

- target dimensions,
- sample count per dimension,
- rank control policy,
- random seed schedule,
- solver backend and tolerance profile,
- witness-threshold policy for numerical cleanup.

Output of a batch is a collection of labeled candidates with metadata. Each candidate receives one of three statuses:

1. rejected: no entanglement evidence survives checks,
2. inconclusive: at least one near-boundary check is unstable,
3. validated-candidate: evidence is stable under all configured checks.

Pipeline A: Randomized PPT Composition + DPS

Pipeline A is the high-throughput screening mechanism.

Algorithm 3.1 (Pipeline A)

Input:

- dimension set D ,
- sample counts N_d for each $d \in D$,
- rank policy R ,
- DPS level/tolerance profile.

Output:

- screened candidate list with DPS-derived evidence.

Procedure:

1. For each $d \in D$, generate N_d random states/maps under rank policy R .
2. Keep only candidates that satisfy PSD and PPT constraints.
3. Convert candidates to the representation required for composition and testing (map or Choi form).
4. Build composed candidates using both self-composition and pairwise composition.
5. Run DPS-based tests and compute witness expectations.
6. Label each case as non-entangled, candidate-entangled, or near-boundary.
7. Export all intermediate certificates and solver logs.

Practical note. Pipeline A is tuned for throughput and broad coverage; therefore its positive hits are treated as preliminary until cross-validated.

Pipeline B: Polynomial/SOS Witness Construction

Pipeline B targets reliability on difficult or ambiguous instances.

Algorithm 3.2 (Pipeline B)

Input:

- candidate family parameters for positive-map construction,
- SOS relaxation level,
- eigenvalue-threshold policy,
- target candidate states from Pipeline A.

Output:

- validated witness candidates and witness-evaluation logs.

Procedure:

1. Construct positive-map candidates using polynomial parameterization.
2. Solve SOS-relaxed SDPs and extract Gram matrices.
3. Perform spectral cleanup: set numerically tiny Gram eigenvalues to zero.
4. Reconstruct cleaned certificates and retest required positivity/non-SOS properties.
5. Apply surviving witnesses to target composed states.
6. Record witness margins and perturbation stability statistics.

Pipeline B is lower-throughput but higher-scrutiny; it is intended to reduce false positives caused by floating-point noise in borderline SDP outcomes [13], [14].

Decision Layer and Cross-Pipeline Validation

The decision layer merges outputs from both pipelines.

Rule set:

1. If Pipeline A rejects and Pipeline B finds no stable witness, label rejected.

2. If either pipeline reports near-zero margins sensitive to tolerance changes, label inconclusive.
3. Label validated-candidate only when both a detection signal and robustness checks are satisfied.

This layer enforces conservative claims: numerical evidence is not promoted to a counterexample without cross-pipeline and stress-test consistency.

Numerical Reliability Protocol

All candidate-entangled outcomes are subjected to a fixed reliability battery.

Protocol 3.3 (Reliability battery)

1. Tolerance sweep: rerun with stricter and looser solver tolerances.
2. Perturbation sweep: apply small random perturbations to candidate data and recompute verdicts.
3. Control sweep: rerun benchmark families in low dimensions where PPT² is known to hold.
4. Consistency sweep: compare witness expectations obtained by independent computation paths.

A candidate fails validation if any sweep changes the sign pattern of critical witness margins in a way inconsistent with robustness.

Reproducibility and Data Logging

Each experiment row stores at least:

- run_id, timestamp, git_commit (if available),
- dimension, rank, seed,
- generator parameters,
- solver name, version, and tolerance settings,
- DPS level and output metrics,
- witness-construction settings and margins,
- final status label and reason code.

This schema supports post-hoc analysis, exact reruns from seeds, and statistical summaries over dimensions and rank classes.

Complexity and Practical Limits

Both pipelines rely on SDPs whose size grows rapidly with dimension and hierarchy level. In practice, this creates three bottlenecks:

1. memory growth in moment/extension matrices,
2. solver instability near degeneracy,
3. long tail runtime for hard instances.

To manage these limits, experiments are organized as parameterized batches with strict logging and intermediate checkpointing.

Experimental Evaluation

Experiments report three groups of results:

- scalability of candidate generation and SDP solving,
- hit rates of candidate-entangled outputs versus rank and dimension,
- stability diagnostics separating robust certificates from numerical artifacts.

Each run logs dimension, rank, random seed, solver configuration, DPS output, witness output, and final status. This enables reproducibility and post-hoc auditing.

Conclusion

This thesis develops a software-first methodology for studying PPT² in high dimensions. The central contribution is a reproducible dual-pipeline framework that combines DPS detection with polynomial/SOS witness construction under explicit numerical reliability controls. Future work includes larger-dimensional searches, improved exact/rational post-certification, and tighter analytical links between map rank and entanglement behavior.

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